

BATO:

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	136636
1	1.00	1.00	1.00	39390
accuracy			1.00	176026
macro avg	1.00	1.00	1.00	176026
weighted avg	1.00	1.00	1.00	176026

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 136636

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 39390

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	0.90	0.98	0.94	136636
1	0.88	0.61	0.72	39390
accuracy			0.89	176026
macro avg	0.89	0.79	0.83	176026
weighted avg	0.89	0.89	0.89	176026

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 133353

False Positives (Normal incorrectly identified as Anomalous): 3283

False Negatives (Anomalous incorrectly identified as Normal): 15241

True Positives (Anomalous correctly identified): 24149

Accuracy: 0.8948

Precision: 0.8803

Recall: 0.6131

F1-Score: 0.7228

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	136636
1	1.00	1.00	1.00	39390
accuracy			1.00	176026
macro avg	1.00	1.00	1.00	176026
weighted avg	1.00	1.00	1.00	176026

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 136636
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 39390
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.98	1.00	0.99	136636
1	0.99	0.95	0.97	39390
accuracy			0.99	176026
macro avg	0.99	0.97	0.98	176026
weighted avg	0.99	0.99	0.99	176026

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 136264
False Positives (Normal incorrectly identified as Anomalous): 372
False Negatives (Anomalous incorrectly identified as Normal): 2112
True Positives (Anomalous correctly identified): 37278
Accuracy: 0.9859
Precision: 0.9901
Recall: 0.9464
F1-Score: 0.9678

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	136636
1	1.00	1.00	1.00	39390

accuracy			1.00	176026
macro avg	1.00	1.00	1.00	176026
weighted avg	1.00	1.00	1.00	176026

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 136636
 False Positives (Normal incorrectly identified as Anomalous): 0
 False Negatives (Anomalous incorrectly identified as Normal): 0
 True Positives (Anomalous correctly identified): 39390
 Accuracy: 1.0000
 Precision: 1.0000
 Recall: 1.0000
 F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	136636
1	1.00	0.99	0.99	39390

accuracy			1.00	176026
macro avg	1.00	1.00	1.00	176026
weighted avg	1.00	1.00	1.00	176026

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 136444
 False Positives (Normal incorrectly identified as Anomalous): 192
 False Negatives (Anomalous incorrectly identified as Normal): 269
 True Positives (Anomalous correctly identified): 39121
 Accuracy: 0.9974
 Precision: 0.9951
 Recall: 0.9932
 F1-Score: 0.9941

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
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0	0.76	0.94	0.84	136636
1	0.00	0.00	0.00	39390

accuracy			0.73	176026
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macro avg	0.38	0.47	0.42	176026
weighted avg	0.59	0.73	0.65	176026

BUSAY:

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\Busay.csv

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	4495
1	1.00	1.00	1.00	198897

accuracy			1.00	203392
macro avg	1.00	1.00	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 4495

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 198897

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
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0	0.98	1.00	0.99	4495
1	1.00	1.00	1.00	198897

accuracy			1.00	203392
macro avg	0.99	1.00	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 4489

False Positives (Normal incorrectly identified as Anomalous): 6

False Negatives (Anomalous incorrectly identified as Normal): 72

True Positives (Anomalous correctly identified): 198825

Accuracy: 0.9996

Precision: 1.0000

Recall: 0.9996

F1-Score: 0.9998

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	4495
1	1.00	1.00	1.00	198897
accuracy			1.00	203392
macro avg	1.00	1.00	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 4495

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 198897

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.98	1.00	0.99	4495
1	1.00	1.00	1.00	198897
accuracy			1.00	203392
macro avg	0.99	1.00	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 4494

False Positives (Normal incorrectly identified as Anomalous): 1

False Negatives (Anomalous incorrectly identified as Normal): 74

True Positives (Anomalous correctly identified): 198823

Accuracy: 0.9996
Precision: 1.0000
Recall: 0.9996
F1-Score: 0.9998

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	4495
1	1.00	1.00	1.00	198897
accuracy			1.00	203392
macro avg	1.00	1.00	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 4495
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 198897
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	0.99	0.99	4495
1	1.00	1.00	1.00	198897
accuracy			1.00	203392
macro avg	1.00	0.99	1.00	203392
weighted avg	1.00	1.00	1.00	203392

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 4448
False Positives (Normal incorrectly identified as Anomalous): 47
False Negatives (Anomalous incorrectly identified as Normal): 10
True Positives (Anomalous correctly identified): 198887
Accuracy: 0.9997
Precision: 0.9998
Recall: 0.9999
F1-Score: 0.9999

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.01	0.36	0.02	4495
1	0.72	0.04	0.07	198897
accuracy			0.04	203392
macro avg	0.36	0.20	0.04	203392
weighted avg	0.70	0.04	0.07	203392

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 1628
False Positives (Normal incorrectly identified as Anomalous): 2867
False Negatives (Anomalous incorrectly identified as Normal): 191618
True Positives (Anomalous correctly identified): 7279
Accuracy: 0.0438
Precision: 0.7174
Recall: 0.0366
F1-Score: 0.0696

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.414150 seconds
Average Jitter: 0.000001 seconds
Throughput: 565.71 bps
Retransmission Rate (estimated): 29.29%

=== Analysis Completed ===

KODIA:

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\kodia_20250414_to_20250421.csv

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	7362
1	1.00	1.00	1.00	168504

accuracy		1.00	175866
macro avg	1.00	1.00	175866
weighted avg	1.00	1.00	175866

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 7362

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 168504

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	0.99	0.99	0.99	7362
1	1.00	1.00	1.00	168504

accuracy		1.00	175866
macro avg	1.00	1.00	175866
weighted avg	1.00	1.00	175866

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 7313

False Positives (Normal incorrectly identified as Anomalous): 49

False Negatives (Anomalous incorrectly identified as Normal): 64

True Positives (Anomalous correctly identified): 168440

Accuracy: 0.9994

Precision: 0.9997

Recall: 0.9996

F1-Score: 0.9997

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	7362
1	1.00	1.00	1.00	168504

accuracy		1.00	175866
macro avg	1.00	1.00	175866
weighted avg	1.00	1.00	175866

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 7362
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 168504
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.99	1.00	1.00	7362
1	1.00	1.00	1.00	168504
accuracy			1.00	175866
macro avg	1.00	1.00	1.00	175866
weighted avg	1.00	1.00	1.00	175866

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 7355
False Positives (Normal incorrectly identified as Anomalous): 7
False Negatives (Anomalous incorrectly identified as Normal): 64
True Positives (Anomalous correctly identified): 168440
Accuracy: 0.9996
Precision: 1.0000
Recall: 0.9996
F1-Score: 0.9998

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	7362
1	1.00	1.00	1.00	168504
accuracy			1.00	175866
macro avg	1.00	1.00	1.00	175866
weighted avg	1.00	1.00	1.00	175866

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 7362
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 168504
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	7362
1	1.00	1.00	1.00	168504
accuracy			1.00	175866
macro avg	1.00	1.00	1.00	175866
weighted avg	1.00	1.00	1.00	175866

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 7351
False Positives (Normal incorrectly identified as Anomalous): 11
False Negatives (Anomalous incorrectly identified as Normal): 3
True Positives (Anomalous correctly identified): 168501
Accuracy: 0.9999
Precision: 0.9999
Recall: 1.0000
F1-Score: 1.0000

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.00	0.06	0.00	7362
1	0.21	0.01	0.02	168504
accuracy			0.01	175866
macro avg	0.11	0.03	0.01	175866
weighted avg	0.20	0.01	0.02	175866

LOREGA:

PS C:\Users\MELBEN\Desktop\Thesis\Anomaly> python matrix.py

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\Lorega.csv

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974
accuracy			1.00	131577
macro avg	1.00	1.00	1.00	131577
weighted avg	1.00	1.00	1.00	131577

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 73603

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 57974

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974
accuracy			1.00	131577
macro avg	1.00	1.00	1.00	131577
weighted avg	1.00	1.00	1.00	131577

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 73603

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 2

True Positives (Anomalous correctly identified): 57972

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974
accuracy			1.00	131577
macro avg	1.00	1.00	1.00	131577
weighted avg	1.00	1.00	1.00	131577

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 73603
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 57974
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974
accuracy			1.00	131577
macro avg	1.00	1.00	1.00	131577
weighted avg	1.00	1.00	1.00	131577

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 73434
False Positives (Normal incorrectly identified as Anomalous): 169
False Negatives (Anomalous incorrectly identified as Normal): 73
True Positives (Anomalous correctly identified): 57901
Accuracy: 0.9982
Precision: 0.9971
Recall: 0.9987
F1-Score: 0.9979

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974

accuracy		1.00	131577
macro avg	1.00	1.00	131577
weighted avg	1.00	1.00	131577

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 73603

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 57974

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	73603
1	1.00	1.00	1.00	57974

accuracy		1.00	131577
macro avg	1.00	1.00	131577
weighted avg	1.00	1.00	131577

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 73603

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 57974

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.56	0.96	0.71	73603
1	0.54	0.06	0.11	57974

accuracy		0.56	131577
macro avg	0.55	0.51	131577
weighted avg	0.56	0.56	131577

PERRELOS:

PS C:\Users\MELBEN\Desktop\Thesis\Anomaly> python matrix.py

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\perrelos_20250404_to_20250411.csv

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	21212
---	------	------	------	-------

1	1.00	1.00	1.00	151293
---	------	------	------	--------

accuracy			1.00	172505
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macro avg	1.00	1.00	1.00	172505
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weighted avg	1.00	1.00	1.00	172505
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Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 21212

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 151293

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.98	0.94	0.96	21212
---	------	------	------	-------

1	0.99	1.00	0.99	151293
---	------	------	------	--------

accuracy			0.99	172505
----------	--	--	------	--------

macro avg	0.98	0.97	0.98	172505
-----------	------	------	------	--------

weighted avg	0.99	0.99	0.99	172505
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Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 19930
False Positives (Normal incorrectly identified as Anomalous): 1282
False Negatives (Anomalous incorrectly identified as Normal): 474
True Positives (Anomalous correctly identified): 150819
Accuracy: 0.9898
Precision: 0.9916
Recall: 0.9969
F1-Score: 0.9942

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	21212
1	1.00	1.00	1.00	151293
accuracy			1.00	172505
macro avg	1.00	1.00	1.00	172505
weighted avg	1.00	1.00	1.00	172505

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 21212
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 151293
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.99	0.96	0.98	21212
1	0.99	1.00	1.00	151293
accuracy			0.99	172505
macro avg	0.99	0.98	0.99	172505
weighted avg	0.99	0.99	0.99	172505

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 20452
False Positives (Normal incorrectly identified as Anomalous): 760
False Negatives (Anomalous incorrectly identified as Normal): 146
True Positives (Anomalous correctly identified): 151147
Accuracy: 0.9947

Precision: 0.9950
Recall: 0.9990
F1-Score: 0.9970

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	21212
1	1.00	1.00	1.00	151293
accuracy			1.00	172505
macro avg	1.00	1.00	1.00	172505
weighted avg	1.00	1.00	1.00	172505

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 21212
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 151293
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	21212
1	1.00	1.00	1.00	151293
accuracy			1.00	172505
macro avg	1.00	1.00	1.00	172505
weighted avg	1.00	1.00	1.00	172505

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 21211
False Positives (Normal incorrectly identified as Anomalous): 1
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 151293
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.08	0.60	0.14	21212
1	0.00	0.00	0.00	151293
accuracy			0.07	172505
macro avg	0.04	0.30	0.07	172505
weighted avg	0.01	0.07	0.02	172505

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 12726
False Positives (Normal incorrectly identified as Anomalous): 8486
False Negatives (Anomalous incorrectly identified as Normal): 151275
True Positives (Anomalous correctly identified): 18
Accuracy: 0.0739
Precision: 0.0021
Recall: 0.0001
F1-Score: 0.0002

SANFER:

PS C:\Users\MELBEN\Desktop\Thesis\Anomaly> python matrix.py

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\SanFernando.csv

===== Anomaly Detection: Power System =====
No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	111347
1	1.00	1.00	1.00	83219
accuracy			1.00	194566
macro avg	1.00	1.00	1.00	194566
weighted avg	1.00	1.00	1.00	194566

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 111347
False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 83219
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	1.00	1.00	1.00	111347
1	1.00	0.99	1.00	83219
accuracy			1.00	194566
macro avg	1.00	1.00	1.00	194566
weighted avg	1.00	1.00	1.00	194566

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 111320
False Positives (Normal incorrectly identified as Anomalous): 27
False Negatives (Anomalous incorrectly identified as Normal): 510
True Positives (Anomalous correctly identified): 82709
Accuracy: 0.9972
Precision: 0.9997
Recall: 0.9939
F1-Score: 0.9968

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	111347
1	1.00	1.00	1.00	83219
accuracy			1.00	194566
macro avg	1.00	1.00	1.00	194566
weighted avg	1.00	1.00	1.00	194566

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 111347
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 83219
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000

F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.97	1.00	0.99	111347
1	1.00	0.96	0.98	83219
accuracy			0.98	194566
macro avg	0.99	0.98	0.98	194566
weighted avg	0.98	0.98	0.98	194566

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 111344

False Positives (Normal incorrectly identified as Anomalous): 3

False Negatives (Anomalous incorrectly identified as Normal): 3296

True Positives (Anomalous correctly identified): 79923

Accuracy: 0.9830

Precision: 1.0000

Recall: 0.9604

F1-Score: 0.9798

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	111347
1	1.00	1.00	1.00	83219
accuracy			1.00	194566
macro avg	1.00	1.00	1.00	194566
weighted avg	1.00	1.00	1.00	194566

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 111347

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 83219

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	111347
1	1.00	1.00	1.00	83219

accuracy			1.00	194566
macro avg	1.00	1.00	1.00	194566
weighted avg	1.00	1.00	1.00	194566

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 111345
 False Positives (Normal incorrectly identified as Anomalous): 2
 False Negatives (Anomalous incorrectly identified as Normal): 1
 True Positives (Anomalous correctly identified): 83218
 Accuracy: 1.0000
 Precision: 1.0000
 Recall: 1.0000
 F1-Score: 1.0000

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.56	0.94	0.70	111347
1	0.27	0.03	0.06	83219

accuracy			0.55	194566
macro avg	0.42	0.48	0.38	194566
weighted avg	0.44	0.55	0.43	194566

TUGAS:

PS C:\Users\MELBEN\Desktop\Thesis\Anomaly> python matrix.py

=== Starting System Analyzer: Full Execution ===

Please enter the full path to your CSV file (used for both analyses):
 C:\Users\MELBEN\Desktop\Thesis\Anomaly\csv file\tugas_20250414_to_20250421.csv

===== Anomaly Detection: Power System =====

No 'Label' column found. Applying automatic labeling based on thresholds.

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1268

1	1.00	1.00	1.00	177451
accuracy			1.00	178719
macro avg	1.00	1.00	1.00	178719
weighted avg	1.00	1.00	1.00	178719

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 1268

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 177451

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.83	0.18	0.30	1268
1	0.99	1.00	1.00	177451

accuracy			0.99	178719
macro avg	0.91	0.59	0.65	178719
weighted avg	0.99	0.99	0.99	178719

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 231

False Positives (Normal incorrectly identified as Anomalous): 1037

False Negatives (Anomalous incorrectly identified as Normal): 48

True Positives (Anomalous correctly identified): 177403

Accuracy: 0.9939

Precision: 0.9942

Recall: 0.9997

F1-Score: 0.9970

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	1.00	1.00	1.00	1268
1	1.00	1.00	1.00	177451

accuracy			1.00	178719
macro avg	1.00	1.00	1.00	178719
weighted avg	1.00	1.00	1.00	178719

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 1268
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 177451
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	0.60	0.20	0.29	1268
1	0.99	1.00	1.00	177451
accuracy			0.99	178719
macro avg	0.80	0.60	0.65	178719
weighted avg	0.99	0.99	0.99	178719

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 248
False Positives (Normal incorrectly identified as Anomalous): 1020
False Negatives (Anomalous incorrectly identified as Normal): 166
True Positives (Anomalous correctly identified): 177285
Accuracy: 0.9934
Precision: 0.9943
Recall: 0.9991
F1-Score: 0.9967

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	1268
1	1.00	1.00	1.00	177451
accuracy			1.00	178719
macro avg	1.00	1.00	1.00	178719
weighted avg	1.00	1.00	1.00	178719

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 1268
False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 177451
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	0.99	0.97	0.98	1268
1	1.00	1.00	1.00	177451
accuracy			1.00	178719
macro avg	0.99	0.98	0.99	178719
weighted avg	1.00	1.00	1.00	178719

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 1228
False Positives (Normal incorrectly identified as Anomalous): 40
False Negatives (Anomalous incorrectly identified as Normal): 14
True Positives (Anomalous correctly identified): 177437
Accuracy: 0.9997
Precision: 0.9998
Recall: 0.9999
F1-Score: 0.9998

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.01	0.97	0.01	1268
1	1.00	0.05	0.10	177451
accuracy			0.06	178719
macro avg	0.50	0.51	0.05	178719
weighted avg	0.99	0.06	0.09	178719

NAGA:

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436

1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 153436
 False Positives (Normal incorrectly identified as Anomalous): 0
 False Negatives (Anomalous incorrectly identified as Normal): 0
 True Positives (Anomalous correctly identified): 203464
 Accuracy: 1.0000
 Precision: 1.0000
 Recall: 1.0000
 F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436
1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 153351
 False Positives (Normal incorrectly identified as Anomalous): 85
 False Negatives (Anomalous incorrectly identified as Normal): 225
 True Positives (Anomalous correctly identified): 203239
 Accuracy: 0.9991
 Precision: 0.9996
 Recall: 0.9989
 F1-Score: 0.9992

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436
1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 153436
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 203464
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436
1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 153285
False Positives (Normal incorrectly identified as Anomalous): 151
False Negatives (Anomalous incorrectly identified as Normal): 480
True Positives (Anomalous correctly identified): 202984
Accuracy: 0.9982
Precision: 0.9993
Recall: 0.9976
F1-Score: 0.9984

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436
1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 153436
False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 203464
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153436
1	1.00	1.00	1.00	203464
accuracy			1.00	356900
macro avg	1.00	1.00	1.00	356900
weighted avg	1.00	1.00	1.00	356900

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 153436
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 14
True Positives (Anomalous correctly identified): 203450
Accuracy: 1.0000
Precision: 1.0000
Recall: 0.9999
F1-Score: 1.0000

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.40	0.89	0.55	153436
1	0.02	0.00	0.00	203464
accuracy			0.38	356900
macro avg	0.21	0.44	0.28	356900
weighted avg	0.19	0.38	0.24	356900

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 136013
False Positives (Normal incorrectly identified as Anomalous): 17423
False Negatives (Anomalous incorrectly identified as Normal): 203042
True Positives (Anomalous correctly identified): 422
Accuracy: 0.3823
Precision: 0.0236
Recall: 0.0021

F1-Score: 0.0038

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.009403 seconds

Average Jitter: 0.000001 seconds

Throughput: 792.55 bps

Retransmission Rate (estimated): 0.93%

=== Analysis Completed ===

TINAGO:

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	1.00	1.00	1.00	163234
---	------	------	------	--------

1	1.00	1.00	1.00	82231
---	------	------	------	-------

accuracy			1.00	245465
----------	--	--	------	--------

macro avg	1.00	1.00	1.00	245465
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weighted avg	1.00	1.00	1.00	245465
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Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 163234

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 82231

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	1.00	1.00	1.00	163234
---	------	------	------	--------

1	1.00	1.00	1.00	82231
---	------	------	------	-------

accuracy			1.00	245465
----------	--	--	------	--------

macro avg	1.00	1.00	1.00	245465
-----------	------	------	------	--------

weighted avg	1.00	1.00	1.00	245465
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Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 163216
False Positives (Normal incorrectly identified as Anomalous): 18
False Negatives (Anomalous incorrectly identified as Normal): 189
True Positives (Anomalous correctly identified): 82042
Accuracy: 0.9992
Precision: 0.9998
Recall: 0.9977
F1-Score: 0.9987

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163234
1	1.00	1.00	1.00	82231
accuracy			1.00	245465
macro avg	1.00	1.00	1.00	245465
weighted avg	1.00	1.00	1.00	245465

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 163234
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 82231
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163234
1	1.00	1.00	1.00	82231
accuracy			1.00	245465
macro avg	1.00	1.00	1.00	245465
weighted avg	1.00	1.00	1.00	245465

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 163234
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 250
True Positives (Anomalous correctly identified): 81981
Accuracy: 0.9990

Precision: 1.0000
Recall: 0.9970
F1-Score: 0.9985

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163234
1	1.00	1.00	1.00	82231
accuracy			1.00	245465
macro avg	1.00	1.00	1.00	245465
weighted avg	1.00	1.00	1.00	245465

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 163234
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 82231
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163234
1	1.00	1.00	1.00	82231
accuracy			1.00	245465
macro avg	1.00	1.00	1.00	245465
weighted avg	1.00	1.00	1.00	245465

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 163187
False Positives (Normal incorrectly identified as Anomalous): 47
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 82231
Accuracy: 0.9998
Precision: 0.9994
Recall: 1.0000
F1-Score: 0.9997

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.68	0.97	0.80	163234
1	0.65	0.10	0.17	82231
accuracy			0.68	245465
macro avg	0.67	0.54	0.49	245465
weighted avg	0.67	0.68	0.59	245465

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 158992
False Positives (Normal incorrectly identified as Anomalous): 4242
False Negatives (Anomalous incorrectly identified as Normal): 74199
True Positives (Anomalous correctly identified): 8032
Accuracy: 0.6804
Precision: 0.6544
Recall: 0.0977
F1-Score: 0.1700

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.047716 seconds
Average Jitter: 0.000007 seconds
Throughput: 763.57 bps
Retransmission Rate (estimated): 4.55%

=== Analysis Completed ===

POBLACION:

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	80208
1	1.00	1.00	1.00	78534
accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 80208
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 78534

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	0.99	1.00	1.00	80208
1	1.00	0.99	1.00	78534
accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 80203

False Positives (Normal incorrectly identified as Anomalous): 5

False Negatives (Anomalous incorrectly identified as Normal): 533

True Positives (Anomalous correctly identified): 78001

Accuracy: 0.9966

Precision: 0.9999

Recall: 0.9932

F1-Score: 0.9966

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	80208
1	1.00	1.00	1.00	78534
accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 80208

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 78534

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	80208
1	1.00	1.00	1.00	78534
accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 80198
False Positives (Normal incorrectly identified as Anomalous): 10
False Negatives (Anomalous incorrectly identified as Normal): 305
True Positives (Anomalous correctly identified): 78229
Accuracy: 0.9980
Precision: 0.9999
Recall: 0.9961
F1-Score: 0.9980

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	80208
1	1.00	1.00	1.00	78534
accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 80208
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 78534
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	1.00	1.00	1.00	80208
1	1.00	1.00	1.00	78534

accuracy			1.00	158742
macro avg	1.00	1.00	1.00	158742
weighted avg	1.00	1.00	1.00	158742

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 80201
 False Positives (Normal incorrectly identified as Anomalous): 7
 False Negatives (Anomalous incorrectly identified as Normal): 2
 True Positives (Anomalous correctly identified): 78532
 Accuracy: 0.9999
 Precision: 0.9999
 Recall: 1.0000
 F1-Score: 0.9999

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

0	0.51	0.96	0.67	80208
1	0.64	0.06	0.12	78534

accuracy			0.52	158742
macro avg	0.58	0.51	0.39	158742
weighted avg	0.58	0.52	0.40	158742

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 77375
 False Positives (Normal incorrectly identified as Anomalous): 2833
 False Negatives (Anomalous incorrectly identified as Normal): 73437
 True Positives (Anomalous correctly identified): 5097
 Accuracy: 0.5195
 Precision: 0.6427
 Recall: 0.0649
 F1-Score: 0.1179

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.164744 seconds
 Average Jitter: 0.000002 seconds
 Throughput: 686.85 bps
 Retransmission Rate (estimated): 14.14%

=== Analysis Completed ===

PUSOK:

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016
accuracy			1.00	169025
macro avg	1.00	1.00	1.00	169025
weighted avg	1.00	1.00	1.00	169025

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 153009

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 16016

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016
accuracy			1.00	169025
macro avg	1.00	1.00	1.00	169025
weighted avg	1.00	1.00	1.00	169025

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 153009

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 74

True Positives (Anomalous correctly identified): 15942

Accuracy: 0.9996

Precision: 1.0000

Recall: 0.9954

F1-Score: 0.9977

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016
accuracy			1.00	169025
macro avg	1.00	1.00	1.00	169025
weighted avg	1.00	1.00	1.00	169025

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 153009
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 16016
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016
accuracy			1.00	169025
macro avg	1.00	1.00	1.00	169025
weighted avg	1.00	1.00	1.00	169025

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 153008
False Positives (Normal incorrectly identified as Anomalous): 1
False Negatives (Anomalous incorrectly identified as Normal): 78
True Positives (Anomalous correctly identified): 15938
Accuracy: 0.9995
Precision: 0.9999
Recall: 0.9951
F1-Score: 0.9975

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016

accuracy		1.00	169025
macro avg	1.00	1.00	169025
weighted avg	1.00	1.00	169025

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 153009

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 16016

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	153009
1	1.00	1.00	1.00	16016

accuracy		1.00	169025
macro avg	1.00	1.00	169025
weighted avg	1.00	1.00	169025

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 153009

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 42

True Positives (Anomalous correctly identified): 15974

Accuracy: 0.9998

Precision: 1.0000

Recall: 0.9974

F1-Score: 0.9987

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.91	0.96	0.93	153009
1	0.22	0.12	0.16	16016

accuracy		0.88	169025
macro avg	0.57	0.54	169025
weighted avg	0.85	0.88	169025

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 146459
False Positives (Normal incorrectly identified as Anomalous): 6550
False Negatives (Anomalous incorrectly identified as Normal): 14118
True Positives (Anomalous correctly identified): 1898
Accuracy: 0.8777
Precision: 0.2247
Recall: 0.1185
F1-Score: 0.1552

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.082018 seconds
Average Jitter: 0.000002 seconds
Throughput: 739.36 bps
Retransmission Rate (estimated): 7.58%

=== Analysis Completed ===

TALAMBAN:

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
0	1.00	1.00	1.00	162615
1	1.00	1.00	1.00	9645
accuracy			1.00	172260
macro avg	1.00	1.00	1.00	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 162615
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 9645
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
0	1.00	1.00	1.00	162615

1	0.98	0.97	0.97	9645
---	------	------	------	------

accuracy			1.00	172260
macro avg	0.99	0.98	0.98	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 162401

False Positives (Normal incorrectly identified as Anomalous): 214

False Negatives (Anomalous incorrectly identified as Normal): 334

True Positives (Anomalous correctly identified): 9311

Accuracy: 0.9968

Precision: 0.9775

Recall: 0.9654

F1-Score: 0.9714

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	162615
1	1.00	1.00	1.00	9645

accuracy			1.00	172260
macro avg	1.00	1.00	1.00	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 162615

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 0

True Positives (Anomalous correctly identified): 9645

Accuracy: 1.0000

Precision: 1.0000

Recall: 1.0000

F1-Score: 1.0000

Training SVM...

Model: SVM

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	162615
1	1.00	0.97	0.99	9645

accuracy			1.00	172260
macro avg	1.00	0.99	0.99	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 162607
False Positives (Normal incorrectly identified as Anomalous): 8
False Negatives (Anomalous incorrectly identified as Normal): 256
True Positives (Anomalous correctly identified): 9389
Accuracy: 0.9985
Precision: 0.9991
Recall: 0.9735
F1-Score: 0.9861

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	162615
1	1.00	1.00	1.00	9645
accuracy			1.00	172260
macro avg	1.00	1.00	1.00	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 162615
False Positives (Normal incorrectly identified as Anomalous): 0
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 9645
Accuracy: 1.0000
Precision: 1.0000
Recall: 1.0000
F1-Score: 1.0000

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	162615
1	1.00	1.00	1.00	9645
accuracy			1.00	172260
macro avg	1.00	1.00	1.00	172260
weighted avg	1.00	1.00	1.00	172260

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 162614
False Positives (Normal incorrectly identified as Anomalous): 1

False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 9645
Accuracy: 1.0000
Precision: 0.9999
Recall: 1.0000
F1-Score: 0.9999

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	0.96	0.96	0.96	162615
1	0.29	0.26	0.28	9645
accuracy			0.92	172260
macro avg	0.63	0.61	0.62	172260
weighted avg	0.92	0.92	0.92	172260

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 156561
False Positives (Normal incorrectly identified as Anomalous): 6054
False Negatives (Anomalous incorrectly identified as Normal): 7116
True Positives (Anomalous correctly identified): 2529
Accuracy: 0.9235
Precision: 0.2947
Recall: 0.2622
F1-Score: 0.2775

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.031542 seconds
Average Jitter: 0.000002 seconds
Throughput: 775.54 bps
Retransmission Rate (estimated): 3.06%

=== Analysis Completed ===

TICAD:

--- Machine Learning Models ---

Training Decision Tree...

Model: Decision Tree

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764

accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for Decision Tree:

True Negatives (Normal correctly identified): 163495
 False Positives (Normal incorrectly identified as Anomalous): 0
 False Negatives (Anomalous incorrectly identified as Normal): 1
 True Positives (Anomalous correctly identified): 1763
 Accuracy: 1.0000
 Precision: 1.0000
 Recall: 0.9994
 F1-Score: 0.9997

Training Logistic Regression...

Model: Logistic Regression

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764

accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for Logistic Regression:

True Negatives (Normal correctly identified): 163495
 False Positives (Normal incorrectly identified as Anomalous): 0
 False Negatives (Anomalous incorrectly identified as Normal): 2
 True Positives (Anomalous correctly identified): 1762
 Accuracy: 1.0000
 Precision: 1.0000
 Recall: 0.9989
 F1-Score: 0.9994

Training Random Forest...

Model: Random Forest

	precision	recall	f1-score	support
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0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764

accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for Random Forest:

True Negatives (Normal correctly identified): 163495

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 1

True Positives (Anomalous correctly identified): 1763

Accuracy: 1.0000

Precision: 1.0000

Recall: 0.9994

F1-Score: 0.9997

Training SVM...

Model: SVM

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764
accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for SVM:

True Negatives (Normal correctly identified): 163495

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 2

True Positives (Anomalous correctly identified): 1762

Accuracy: 1.0000

Precision: 1.0000

Recall: 0.9989

F1-Score: 0.9994

Training Gradient Boosting...

Model: Gradient Boosting

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764
accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for Gradient Boosting:

True Negatives (Normal correctly identified): 163495

False Positives (Normal incorrectly identified as Anomalous): 0

False Negatives (Anomalous incorrectly identified as Normal): 1

True Positives (Anomalous correctly identified): 1763

Accuracy: 1.0000
Precision: 1.0000
Recall: 0.9994
F1-Score: 0.9997

Training MLP Classifier...

Model: MLP Classifier

	precision	recall	f1-score	support
0	1.00	1.00	1.00	163495
1	1.00	1.00	1.00	1764
accuracy			1.00	165259
macro avg	1.00	1.00	1.00	165259
weighted avg	1.00	1.00	1.00	165259

Detailed Metrics for MLP Classifier:

True Negatives (Normal correctly identified): 163494
False Positives (Normal incorrectly identified as Anomalous): 1
False Negatives (Anomalous incorrectly identified as Normal): 2
True Positives (Anomalous correctly identified): 1762
Accuracy: 1.0000
Precision: 0.9994
Recall: 0.9989
F1-Score: 0.9991

--- Autoencoder Alternative (PCA) ---

PCA-based Anomaly Detection Results:

	precision	recall	f1-score	support
0	1.00	0.96	0.98	163495
1	0.21	1.00	0.35	1764
accuracy			0.96	165259
macro avg	0.61	0.98	0.67	165259
weighted avg	0.99	0.96	0.97	165259

Detailed Metrics for PCA-based Detection:

True Negatives (Normal correctly identified): 156998
False Positives (Normal incorrectly identified as Anomalous): 6497
False Negatives (Anomalous incorrectly identified as Normal): 0
True Positives (Anomalous correctly identified): 1764
Accuracy: 0.9607
Precision: 0.2135
Recall: 1.0000
F1-Score: 0.3519

===== Network Performance Analysis =====

--- Network Metrics ---

Average Latency: 1.149500 seconds

Average Jitter: 0.000002 seconds

Throughput: 695.95 bps

Retransmission Rate (estimated): 13.01%

=== Analysis Completed ===